

Digital Payments Fraud Detection Case Study

1. Background

A leading digital payments platform processes millions of transactions daily across multiple channels. With rapid growth in online payments, the company has observed an increase in fraudulent activities, leading to financial losses, regulatory concerns, and declining customer trust.

2. Business Problem

Recent internal reports indicate that existing rule-based fraud detection systems are failing to effectively capture evolving fraud patterns. Leadership is concerned about rising fraud losses and inefficiencies in current monitoring systems.

3. Analytical Objective

The objective is to analyze transaction data (~5.5M records) to determine key fraud patterns, identify high-risk transactions, and support the development of a scalable fraud detection strategy.

4. Key Hypotheses to Investigate

- Fraud is concentrated in specific transaction types such as TRANSFER and CASH_OUT
- High-value transactions have a higher probability of fraud
- Sudden balance changes indicate suspicious activity
- Current system may generate excessive false positives

5. Dataset Overview

The dataset includes transaction-level details such as type, amount, account balances, and fraud labels. The dataset is highly imbalanced, with a small percentage of fraudulent transactions.

6. Analysis Plan

- Perform data cleaning and validation
- Conduct exploratory data analysis (EDA)
- Segment transactions by type, amount, and behavior
- Identify fraud patterns and anomalies

7. Key Analysis Areas

- Fraud distribution across transaction types
- Fraud probability by transaction amount
- Balance change patterns
- Identification of high-risk transaction thresholds

8. Expected Outcome

The analysis will provide a clear understanding of fraud drivers, risk segments, and behavioral patterns. It will support data-driven decision-making for fraud prevention strategies.

9. Business Impact

- Reduction in fraud-related losses - Improved fraud detection efficiency - Better customer experience by reducing false alerts

10. Strategic Recommendations

- Implement machine learning-based fraud detection - Monitor high-risk transaction categories - Set dynamic alert thresholds - Continuously update fraud detection models